

Venkata Sitha Madhav Vootla

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EDUCATION

University at Buffalo, The State University of New York

Master of Science in Computer Science

Aug 2023 – Dec 2024

Buffalo, New York

Vellore Institute of Technology

Bachelor of Technology in Electronics and Communication Engineering

Aug 2017 – May 2021

Vellore, India

EXPERIENCE

Wipro LTD

Project Engineer

July 2021 – July 2023

- Worked with **Dell Technologies** on the Corruption and Risk Detection System (CARDS) project, leveraging **Angular**, **.NET**, **MongoDB**, and **Microsoft SQL Server** to develop web application that enhanced fraud detection and monitored sales across 5 business units.
- Designed and developed a high-performance, **microservices** based application following **SOLID** principles, implementing **RESTful APIs** with **.NET Core** and a dynamic UI using **Angular**, **HTML**, **CSS**, **TypeScript**.
- Optimized **database schemas** and **tables** using Microsoft SQL Server and MongoDB, utilizing **ADO.NET** and **Entity Framework** for efficient data access, improved query performance and scalability.
- Developed and optimized **stored procedures** and **SQL queries** for handling large volumes of data, reducing query execution times, resulting in a **20%** reduction in application's loading time.
- Implemented **JWT authentication** and **Role-Based Access Control (RBAC)** in .NET Web API, enhancing security and ensuring authorized access to API endpoints based on user roles and permissions.
- Identified and resolved over **20 critical bugs** in a single sprint contributing to a **15%** improvement in overall efficiency of application.
- Collaborated with **DevOps** teams for managing **source control** with **Git** and **GitLab CI/CD pipelines** for deploying the web application across **dev**, **sit**, and **prod** environments within the **VMware on-premise** infrastructure to ensure smooth transitions from development to production.
- Developed **end-to-end** functionality of a module, including **API endpoints**, implementing **Figma** designs in UI that played a pivotal role in identifying **50+** fraudulent sales representatives and **\$150K** in fraudulent sales.
- Presented **Sprint** reviews and demonstrations of new features to business heads and **end customers** as part of the **Agile Scrum** process, gathering direct feedback to ensure alignment with business objectives.
- **Migrated** a legacy application to **cloud-native** architectures using AWS services (**ECS**, **RDS**, **S3**), containerized with **Docker**, and orchestrated with **Kubernetes**, enhancing scalability and performance.
- **Promoted** to a **higher career band** within one year in the organization due to demonstrated expertise in code optimization and timely deliverable releases, consistently exceeding project expectations.

PROJECTS

Movie Ticket Booking System

April 2021 – July 2021

- Designed and developed backend **RESTful API** endpoints using the **.NET** Framework, implementing efficient server-side logic providing real-time availability of seats, screens, and role-based authentication for users.
- Designed, optimized, and managed a PostgreSQL database, leveraging ADO.NET, Entity Framework, and **LINQ** to establish a robust data access layer for performing **CRUD** operations.
- Integrated **Swagger** for API documentation and used **Postman** for API testing and validation, enabling easy exploration of API endpoints and ensuring proper functionality and performance before deployment.
- Deployed and hosted the application on **AWS EC2**, configuring instances, optimizing server performance, and ensuring high availability and reliability in a cloud environment.

TECHNICAL SKILLS

Programming Languages : C#, Java, Python, TypeScript, JavaScript, SQL, HTML, CSS

Frameworks : Angular, .NET

Databases : Microsoft SQL Server, PostgreSQL, MongoDB

.NET Frameworks : ASP.NET MVC, ASP.NET Web API, ADO.NET, Entity Framework

Containerization : Docker, Kubernetes, Microservices

IDEs : Visual Studio, VS Code, MS SQL Server Management Studio, IntelliJ IDEA, Jupyter Notebook

Tools : Git, GitLab, CI/CD, Jira, Confluence, Figma, Postman, Swagger, VMware, AWS